Fewer Fires, Bigger Burns

Where Is Canada's Burned Area Coming From?

COMP 6934 – Final Project Report

Team 9

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1 Introduction

Our data is the fire point file of the *Canadian National Fire Database* (NFDB), published by Natural Resources Canada under the Open Government Licence – Canada [1]. We use the version dated 11 August 2026. Each row is one wildfire reported by a province, a territory or Parks Canada. The file has 448,618 fires going back to the 1940s and 24 columns. For every fire we know where it started (latitude and longitude), which agency reported it, the date it was reported, its final size in hectares and its cause, which is coded as human (H), natural/lightning (N) or unknown (U). The file also has an "out" date, a fire type and a few agency-specific columns, but these are filled in very unevenly.

We picked this dataset because of the last few fire seasons. In 2023 Canada had its worst season on record: the fires in this file add up to 17.6 million hectares, more than twice the previous record from 1989. 2025 was the second worst year at 8.7 million hectares. Wildfires matter for forestry, for people's health through smoke, for emergency planning, and especially for Indigenous and northern communities that get evacuated. As students living in Canada we hear about smoke and evacuations every summer, and we wanted to see what the data says about these record years.

2 Questions

The headline numbers seem to go in opposite directions: Canada reports fewer fires than it used to, but more land is burning. Our main question is therefore:

Where is Canada's burned area coming from?

The NFDB has no weather or climate information, so we cannot say *why* fires are getting bigger. What we can do is describe which fires make up the burned area, in terms of their size, cause, location and start date. We split the main question into five smaller ones:

1. How have the number of fires and the area burned changed since 1980, and how unusual were 2023 and 2025? Is the drop in fire numbers real, or a change in reporting?
2. Does most of the burned area come from a few very big fires?
3. How do human-caused and lightning-caused fires differ in number and in area burned?
4. Where do the big fires happen?
5. When in the year do fires start, and did 2023 and 2025 start earlier than usual?

3 Data preparation

We keep the years 1980 to 2025. Before that the record is not complete: Ontario and Quebec only start reporting in the mid-1970s, and Saskatchewan in 1980. All cleaning is done in one Python script (`make_figures.py`), which also makes every figure in this report. Table 1 lists what we did with each problem we found.

Problem	What we did
Years before 1980, or −999	Removed (108,439 rows). 340,179 fires remain.
Prescribed burns	Removed (438). They are planned fires, not wildfires. Agencies flag them in different columns, so we checked all of them.
Exact repeated rows	Removed 1,201 rows that have the same agency, date, location, size and cause (82,900 ha in total). The fire ID alone is not unique, because some agencies reuse IDs for different fires.
Positive longitudes	83 fires (mostly Newfoundland and Labrador, 2022–25) are missing the minus sign. We flipped the sign.
Impossible locations	117 fires are still outside Canada (e.g. at 0, 0). Kept in all totals, left off the map.
Fires of 0 ha	29,525 fires. Kept, since they are real reports; only left out of the log-scale histogram.
Unknown start month (0)	3,459 fires, mostly Manitoba 2000–2003. Left out of the month heatmap only.
No "out" date	37.8% of fires. We do not use this column.

Table 1: Data problems and how we handled them. After cleaning, 338,540 fires from 1980–2025 remain.

4 Results

4.1 Fewer fires, but more area burned

Figure 1 shows the starting point of our project. In the 1980s agencies reported on average 8,691 fires a year; in the last ten years (2016–25) the average is 5,716, a drop of 34%. Over the same periods the average area burned went up from 2.7 to 4.7 million hectares (Mha) a year. The area line is also much more "spiky" than the count line. A typical year burns about 1.9 Mha (the median), and a few years stand far above it: 1989, 1995, 2025 and above all 2023. With 17.6 Mha, 2023 burned about nine times a typical year and 2.3 times the old record of 1989. 2025 is the second largest year at 8.7 Mha. What stands out is that both of these years had an ordinary number of fires (6,837 and 6,215). The record area did not come from more fires.

Before we trust the drop in fire numbers, we have to check that it is not a reporting effect, for example one agency that stopped sending small fires. Figure 2 splits the counts by agency. Nine agencies have full records since 1980, and seven of them report fewer fires now than in the 1980s: Ontario (−59%), Quebec (−51%), Manitoba (−47%), British Columbia (−43%), the Northwest Territories (−43%), Saskatchewan (−39%) and Yukon (−19%). Only Alberta (+14%) and Parks Canada, which is small and whose counts jump around 2000, go up. Taken together, these nine agencies went from 8,427 to 5,157 fires a year (−39%), slightly more than the national drop. New Brunswick, Nova Scotia and Newfoundland and Labrador only send full records from the late 1980s, so they push the national total *up* rather than down. Since the decline is spread across almost every agency, we treat it as real. We still

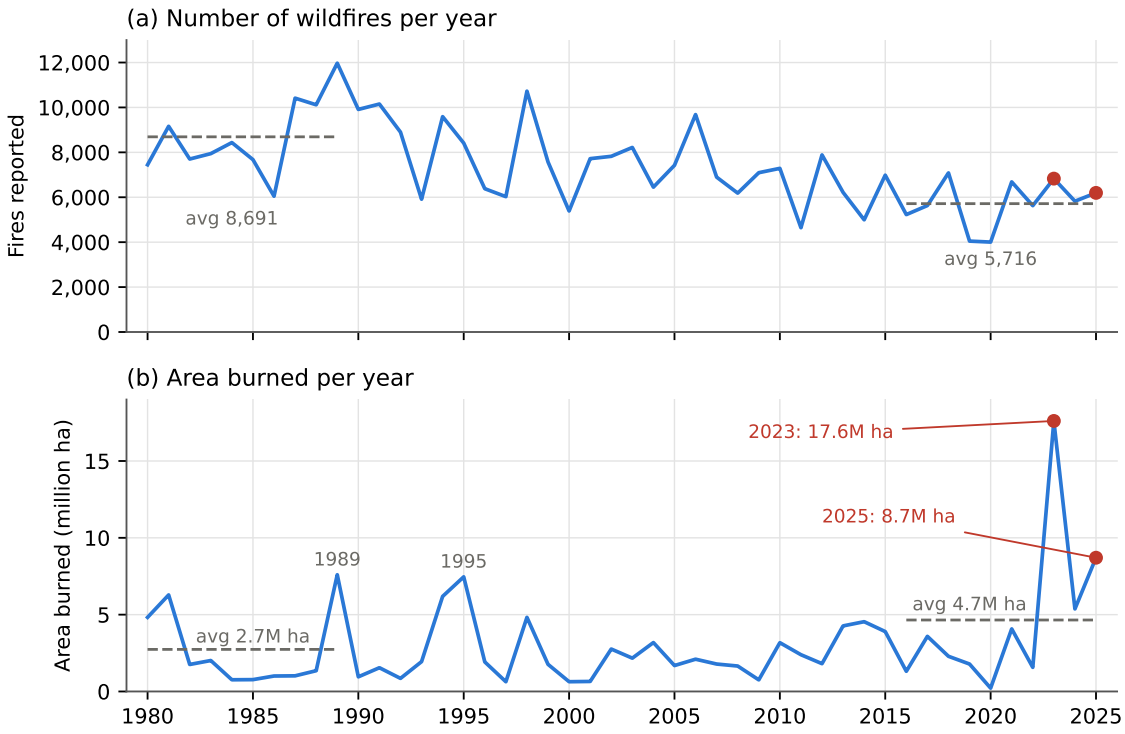


Figure 1: Number of wildfires (a) and area burned (b) per year, 1980–2025. Dashed lines are the averages for 1980–89 and 2016–25. Fires went down by about a third while the average area burned went up by about 70%, and 2023 (17.6 Mha) is far above every other year.

cannot rule out that some agencies became stricter about which small fires they record, but there is no single break that would point to that.

4.2 A few very large fires make up almost all of the area

Figure 3a shows how many fires there are of each size. Most fires are tiny: 71.5% end below one hectare, and the most common recorded size is 0.1 ha. The right tail is long, with fires up to 1.05 million hectares. Figure 3b uses the same x-axis to show the running total of area burned. The curve stays flat at zero across all the small fires and only starts to rise after a few hundred hectares. The 15,914 fires above 200 ha are only 4.7% of all fires, but they burned 99.0% of the area. Going further, the 2,613 fires of at least 10,000 ha (0.8% of fires) account for 80.2% of all area burned.

This also explains figure 1. While the total number of fires fell, the number of fires above 200 ha went *up*, from 334 a year in the 1980s to 437 a year in 2016–25 (+31%). 2023 had 953 such fires; the previous highest was 751 in 1989. In 2023, 91% of the area burned came from fires of 10,000 ha or more. So the answer to the second question is clearly yes: Canada's burned area is the area of its few very large fires, and the small fires, even though they are most of the work for fire crews, hardly change the total.

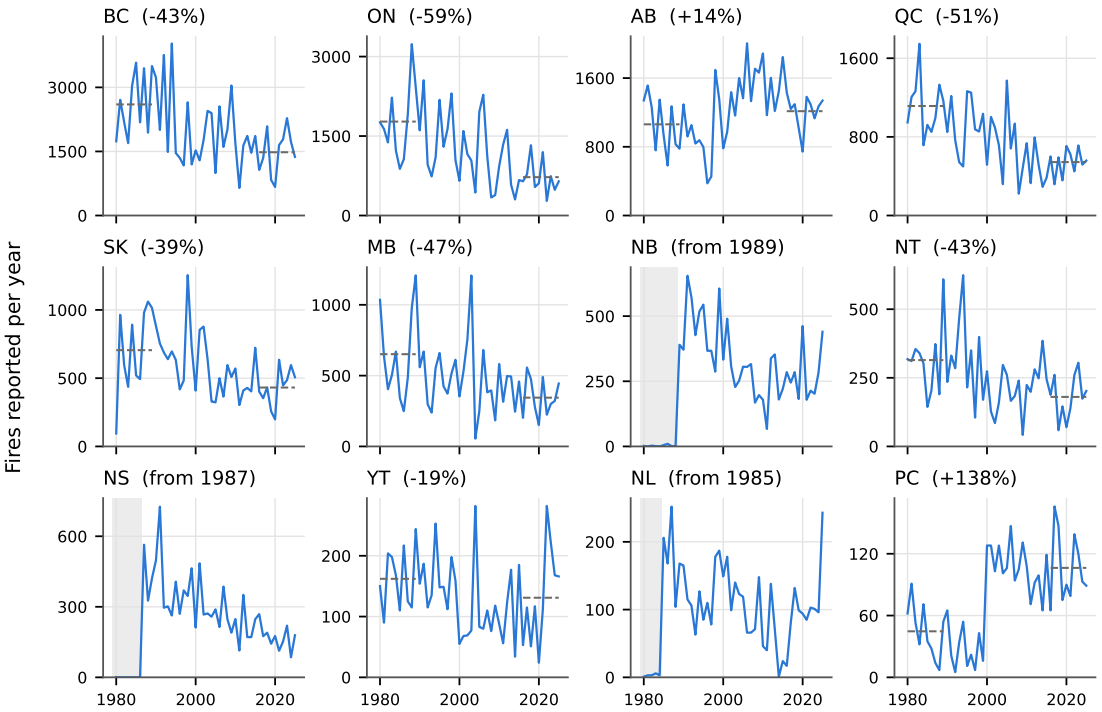


Figure 2: Fires reported per year by each agency (each panel has its own y-axis; PEI is left out, it has only 55 fires). The percentage is the change between the 1980–89 and 2016–25 averages (dashed). Grey areas mark years before NB, NS and NL sent full records. The drop in figure 1a shows up in almost every agency, so it is not caused by one agency changing how it reports.

4.3 Lightning fires are half the fires but almost all the area

Figure 4 compares the two main causes. People start 51.3% of all fires, a little more than lightning (45.7%), but human-caused fires burn only 8.8% of the area. Lightning fires burn 87.5%. The typical fire is small for both causes (median 0.1 ha for human and 0.2 ha for lightning), so the difference is all in the tail: 83% of the fires above 200 ha were started by lightning. In 2023 the lightning share of the area was even higher, 92%. A likely reason, which we cannot test with this data, is that human fires start close to roads and towns where they are seen early and attacked quickly, while lightning fires often start in remote forest where they can grow for days. The next figure supports the location part of this explanation.

4.4 The big fires are in the northern boreal forest

Figure 5 maps the 15,914 fires above 200 ha (province outlines from Natural Earth [2]). They are not spread evenly. Most of them form a wide band through the boreal forest: from Yukon and the Northwest Territories, across northern Alberta, Saskatchewan and Manitoba, into northern Ontario and Quebec. 68.8% of their area lies north of 55°N, where few people live. The agencies with the most large-fire area are the Northwest Territories (22%) and Saskatchewan (19.5%), followed by Manitoba, Quebec, Alberta and British Columbia (8–12% each). Ontario and the Atlantic provinces have relatively few large fires, despite Ontario's

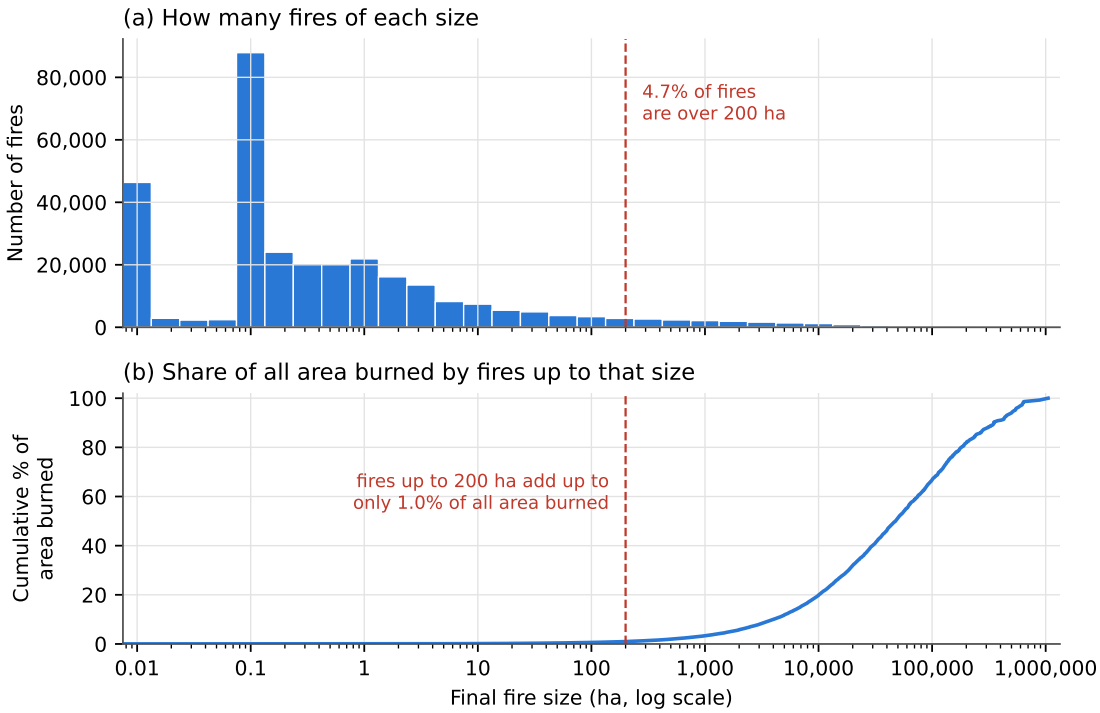


Figure 3: (a) Histogram of final fire size on a log scale (fires of 0 ha left out). The spikes at 0.01 and 0.1 ha are sizes rounded by the agencies. (b) Share of all area burned 1980–2025 that comes from fires up to a given size. Fires larger than 200 ha are 4.7% of all fires but 99% of the area, so the burned area is decided by the few very large fires.

high fire count in figure 2. The orange dots (human cause) cluster along the southern edge of the forest, near farmland, roads and towns in central Alberta, Saskatchewan and Manitoba and in southern BC; 58% of the large human-caused fire area is south of 55°N. This matches the idea from the previous section: the largest fires start where nobody is around to stop them early.

4.5 2023 and 2025 started early

Figure 6 looks at the timing of the season. Each column is one year, and the colour shows what share of that year's area came from fires starting in each month. The darkest cells in most years are in June and July: these are the months of the lightning season. In a typical year (the median), only 15% of the area comes from fires that started by the end of May. In 2023 this share was 37% and in 2025 it was 49%. In absolute terms, fires that started in May 2023 burned 6.5 Mha, more than the whole of any other year in the record except 1989, 1995 and 2025. In May 2025 it was 4.3 Mha, while a typical May adds only about 0.2 Mha.

The heatmap also shows that an early start alone does not make a record year. In 1980, 2016 and 2019 about half of the area also came from fires that started by the end of May, but these years burned only 1.3 to 4.8 Mha in total. What made 2023 different is that the early fires came on top of a very active June and July: fires starting in June and July 2023 burned 4.8 and 5.0 Mha, each of these alone more than most full years. So 2023 had a season

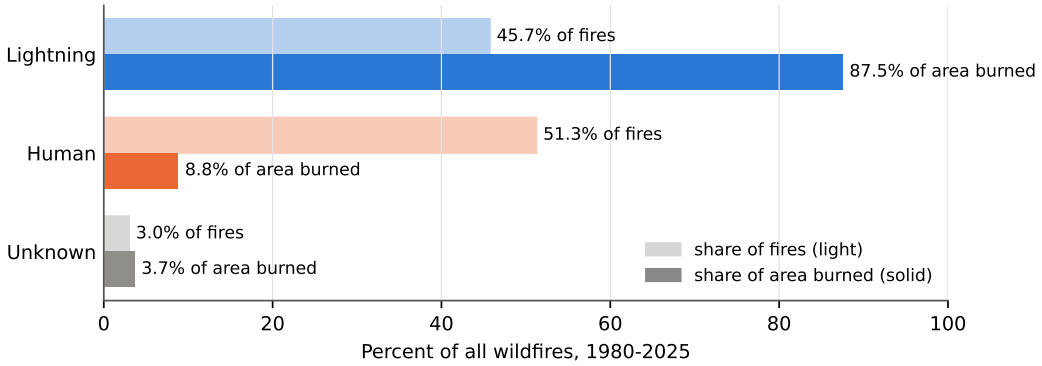


Figure 4: Share of all fires (light bars) and share of all area burned (solid bars) for each cause, 1980–2025. Humans start slightly more than half of the fires but these burn under 9% of the area, while lightning fires burn 87.5%.

that started early *and* stayed active.

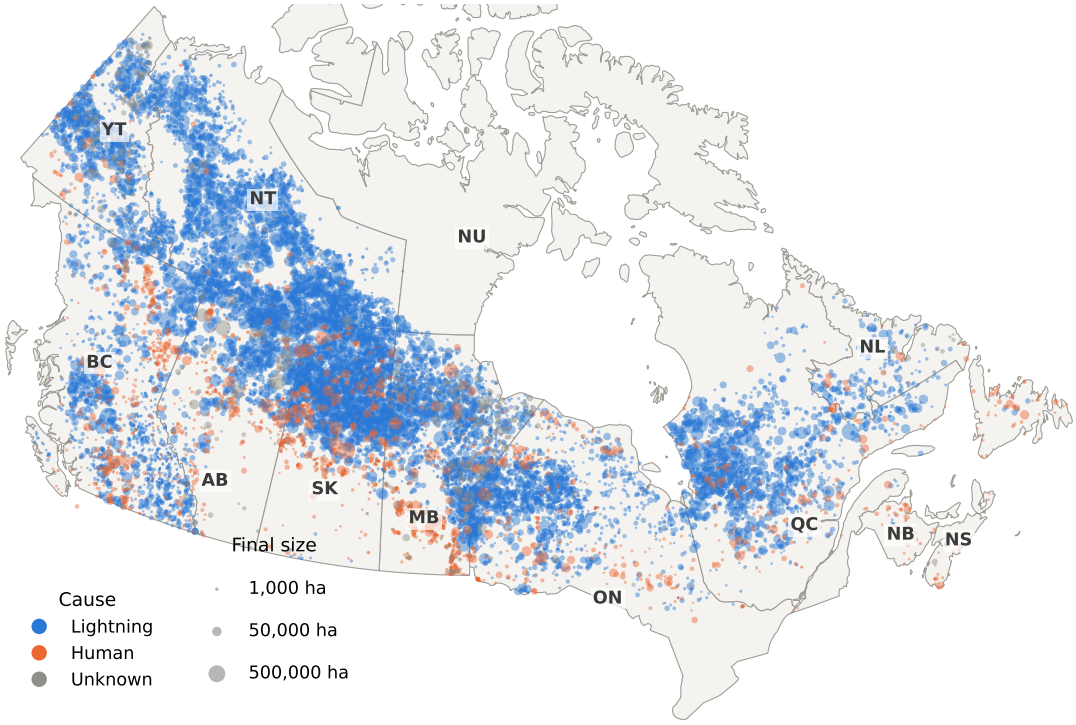


Figure 5: Start locations of all fires larger than 200 ha, 1980–2025. Dot area is proportional to final fire size and colour shows the cause. Large fires form a band through the northern boreal forest from Yukon to Quebec, mostly started by lightning; human-caused large fires sit along the southern edge of that band, close to where people live.

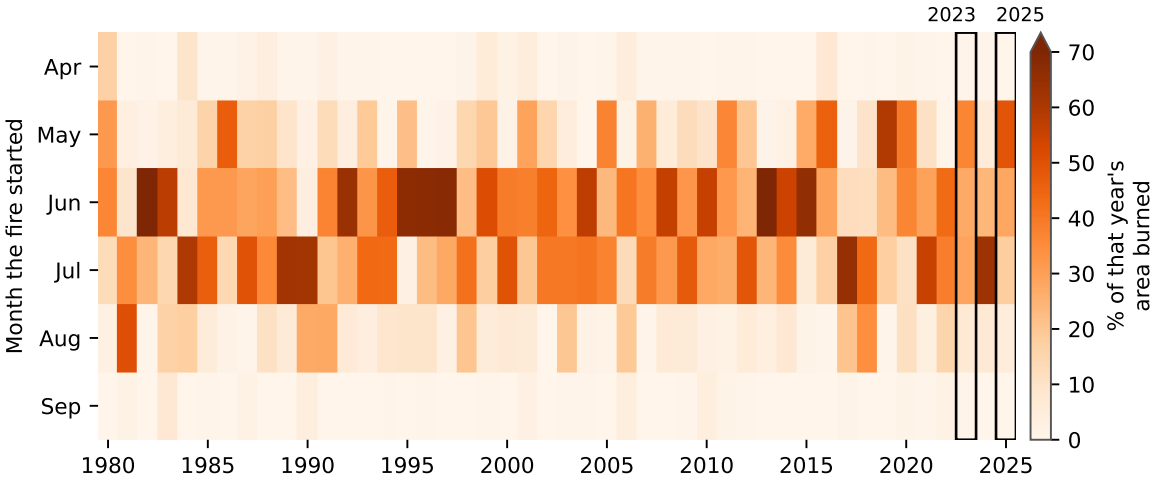


Figure 6: Heatmap of how each year's area burned splits by the month the fires started (each column sums to about 100%; fires starting October to March make up only 0.5% of all area and are left out). In most years the area comes from fires that start in June and July. In 2023 and 2025 (boxed) a much larger share came from fires that started in May.

5 Conclusions

We started with an apparent contradiction: fewer fires, but more area burned. Our figures resolve it.

- **The drop in fire numbers looks real.** Fires fell by about a third since the 1980s, in seven of the nine agencies with complete records (figures 1 and 2).
- **The burned area comes from very few fires.** Fires over 200 ha are under 5% of fires but 99% of the area, and their number went up by 31% while the total went down (figure 3). This is why the two lines in figure 1 move apart.
- **Those fires are mostly lightning fires in the north.** Lightning starts under half of the fires but burns 87.5% of the area (figure 4), mostly north of 55°N (figure 5).
- **Record years start early and stay active.** 2023 and 2025 got a much bigger share of their area from May fires, and in 2023 June and July were also very large (figure 6).

So, to answer our question: Canada's burned area comes from a small number of very large, mostly lightning-caused fires in the remote northern forest, and in the worst years from fires that started unusually early in the season. Human-caused fires are more numerous, but they are smaller and closer to people, so they add little to the total.

Limitations. The NFDB has no weather, fuel or fire-suppression data, so our results describe the burned area but cannot explain the change. The data are points: each fire's whole area is placed at its start location, which is fine for large-scale patterns but not for exact fire footprints. Agencies also differ in how they record size, cause and dates, and small fires are often rounded to 0.1 ha. Joining the NFDB with fire-weather data, or using the NFDB fire polygons, would be good next steps.

Code. All figures and numbers in this report are produced by `make_figures.py` from the 11 August 2026 copy of the data.

Contributions. All three team members contributed equally to this project.

References

[1] Canadian Forest Service. *Canadian National Fire Database – Agency Fire Data*, fire point file, version 20260811. Natural Resources Canada, Canadian Forest Service, Northern Forestry Centre, Edmonton, Alberta. `https://cwfis.cfs.nrcan.gc.ca/ha/nfdb`

[2] Natural Earth. *Admin 1 – States, Provinces*, 1:50m, version 5.1.1. `https://www.naturalearthdata.com`